

CHORNOMORS'KE, UkraineWMO: **339240**Lat: **45.5217N**Lon: **32.7161E**Elev: **10**StdP: **101.20**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.7	-8.0	-13.9	1.1	-9.6	-11.1	1.5	-7.0	16.2	-0.8	14.1	-2.0	5.5	70	0.523

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	30.1	22.4	28.7	22.1	27.6	21.8	24.2	27.9	23.4	27.2	22.6	26.5	5.2	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	17.7	27.1	22.0	16.7	26.4	21.2	15.9	25.5	73.1	27.9	69.8	27.1	66.9	26.7	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.8	9.9	-13.5	34.0	3.9	2.4	-16.3	35.7	-18.7	37.1	-20.9	38.5	-23.7	40.2		
(5)			-13.8	25.4	3.8	1.2	-16.6	26.3	-18.8	27.0	-21.0	27.7	-23.8	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.9	1.2	1.7	5.0	10.0	15.6	20.5	23.2	22.9	18.2	12.3	7.7	3.7
	DBStd	8.64	4.91	4.94	3.54	3.20	3.12	2.50	2.17	2.46	2.96	3.99	4.33	4.92
	HDD10.0	1020	273	232	158	39	1	0	0	0	23	93	201	
	HDD18.3	2764	531	464	413	251	96	8	0	1	38	188	318	455
	CDD10.0	1713	1	1	3	38	174	315	409	401	245	96	26	5
	CDD18.3	414	0	0	0	0	10	73	152	143	33	3	0	0
	CDH23.3	2562	0	0	0	1	35	321	1026	1049	125	5	0	0
	CDH26.7	491	0	0	0	0	3	42	198	234	12	1	0	0
(14) Wind	WSAvg	4.4	5.1	5.2	5.2	4.3	3.6	3.7	3.8	3.8	4.1	4.5	4.9	5.1
(15) Precipitation	PrecAvg	399	33	32	25	28	25	36	34	37	42	30	36	45
	PrecMax	581	84	77	53	71	69	136	120	170	173	106	119	104
	PrecMin	193	2	7	2	1	0	1	0	0	0	2	2	6
	PrecStd	100	23	16	15	18	20	29	36	41	42	25	24	26
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.9	13.3	15.9	21.4	26.3	30.0	32.6	33.1	28.2	23.5	19.1	15.0
		MCWB	9.5	9.6	10.4	13.8	17.8	21.3	21.6	21.6	19.0	18.2	15.1	11.8
	2%	DB	10.2	11.1	13.1	18.5	24.0	27.8	30.0	30.5	26.2	21.3	17.1	13.1
		MCWB	8.4	8.5	9.1	12.4	18.2	21.2	22.2	22.9	19.5	17.3	14.1	11.1
	5%	DB	9.0	9.5	11.3	16.5	22.3	26.2	28.6	29.0	24.7	19.6	15.4	11.6
		MCWB	7.6	7.6	8.2	11.8	17.4	20.8	22.5	22.3	19.3	16.4	13.1	10.0
	10%	DB	7.9	8.1	9.8	15.0	20.9	25.1	27.5	27.8	23.5	18.2	13.9	10.4
		MCWB	6.6	6.7	7.4	11.0	16.7	20.3	22.1	21.9	18.5	15.4	12.1	8.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	10.1	10.8	14.9	20.4	23.6	25.0	26.2	22.5	19.3	15.8	12.8
		MCDB	11.5	12.2	14.8	20.3	23.8	27.3	28.4	29.7	25.6	22.1	18.2	14.4
	2%	WB	8.6	9.0	9.5	13.4	19.2	22.5	24.2	24.5	21.1	18.1	14.4	11.2
		MCDB	9.9	10.7	12.3	17.2	22.4	25.9	27.7	28.4	24.4	20.5	16.5	12.7
	5%	WB	7.7	7.7	8.6	12.3	18.2	21.7	23.5	23.5	20.2	16.9	13.4	10.1
		MCDB	8.9	9.2	10.9	15.4	21.3	25.1	27.2	27.6	23.3	19.3	15.1	11.6
	10%	WB	6.6	6.7	7.7	11.5	17.2	21.0	22.8	22.7	19.2	15.7	12.3	9.0
		MCDB	7.8	8.1	9.6	14.4	20.2	24.4	26.7	26.7	22.4	17.8	13.9	10.3
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.3	5.5	7.0	7.2	6.7	6.9	7.5	7.4	6.8	5.7	4.8
		MCDBR	5.9	6.8	8.0	9.6	9.8	8.8	9.1	9.9	9.6	8.1	7.0	5.4
	5% WB	MCWBR	4.7	4.8	4.8	5.2	5.1	4.6	4.8	5.3	5.3	5.1	4.8	4.4
		MCWBR	5.8	6.2	7.2	8.4	8.3	7.5	7.7	8.1	8.0	7.1	6.5	5.2
(40) Clear-Sky Solar Irradiance	taub		0.350	0.376	0.401	0.439	0.417	0.418	0.438	0.442	0.413	0.381	0.362	0.348
		taud	2.327	2.257	2.219	2.139	2.300	2.342	2.292	2.263	2.312	2.372	2.343	2.328
	Ebn at Noon		735	785	820	820	853	852	829	809	800	774	713	691
		Edh at Noon	80	103	122	144	127	122	127	126	111	91	78	73
(44) All-Sky Solar Radiation	RadAvg		1.10	2.00	3.33	5.08	6.40	6.77	6.79	5.93	4.32	2.67	1.39	0.91
	RadStd		0.16	0.24	0.32	0.38	0.40	0.39	0.29	0.33	0.40	0.26	0.16	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.62	N/A	N/A	+0.80	N/A	N/A	-119	-158	+128	+79
Regional (6 neighbors)	+0.67	N/A	N/A	+0.91	+0.43	+0.37	-110	-143	+121	+78